

Why Language Models Hallucinate: A Statistical Account

A Summary and Synthesis of Current Research | Compiled 2026

ABSTRACT

Hallucination — the tendency of large language models (LLMs) to produce fluent, confident, and factually wrong statements — is often described as mysterious or unpredictable. Recent theoretical work, most notably by Kalai, Nachum, Vempala, and Zhang (2025), argues the opposite: hallucination is a natural and largely predictable consequence of how models are trained and graded, not an unexplained quirk of scale. This document synthesizes that argument alongside supporting survey literature, covering the pretraining origins of factual error, why post-training and benchmarking fail to eliminate it, the specific error categories that resist correction regardless of model size, and the mitigation strategies — including retrieval-augmented generation and evaluation reform — currently under study.

1. Framing the Problem

Hallucination is usually discussed as a mysterious failure mode — as if the model occasionally and inexplicably loses its grip on truth. The 2025 OpenAI/Georgia Tech analysis reframes this: a language model producing a plausible but false statement under uncertainty behaves the same way a student does when guessing on a hard exam question rather than leaving it blank. The core claim of that paper is that this behavior is not accidental — it is actively rewarded by the training and evaluation procedures that dominate the field.

2. Pretraining: Hallucination as a Classification Error

The paper's central technical move is to reduce next-token generation to a binary classification problem: given a candidate statement, is it valid or not? Framed this way, hallucinations stop looking mysterious and start looking like ordinary statistical classification errors. If a model cannot reliably distinguish correct statements from incorrect ones during training, errors in generation follow as a direct, provable consequence — independent of architecture, scale, or how carefully the model was engineered.

2.1 Why Some Facts Are Unavoidably Guessed

One specific and easy-to-verify driver is what the authors call the singleton rate: the proportion of facts that appear only once in the training corpus. A fact seen once gives the model no repeated statistical signal to anchor on — there is no pattern to learn, only a single, arbitrary data point (an example commonly cited is an individual's exact birthday). At least that fraction of such queries will be answered incorrectly by any model, no matter how well calibrated, because there is nothing in the data distribution to learn from.

2.2 Garbage In, Garbage Out (GIGO)

Training corpora are not error-free. Corpora scraped from the web contain factual mistakes, outdated claims, and fabrications, and base models reproduce these errors because they have no independent mechanism for verifying the content they are trained on. This compounds the classification-error problem described above rather than replacing it.

2.3 Structural and Distributional Limits

Beyond data-driven causes, several factors are rooted in the model or the task itself, and do not shrink simply by adding more parameters or more data:

- Poor-model errors: tasks such as counting characters or tokens are mismatched to how subword-based models represent text, producing errors that are structural rather than statistical.
- Out-of-distribution (OOD) inputs: queries far outside the training distribution (for example, a heavily English-trained model facing a complex query in a low-resource language) leave the model without reliable statistical anchors.
- Computational hardness: some tasks (such as decrypting ciphertext without a key) are provably hard regardless of model competence; a calibrated model must fail on them rather than guess convincingly.

3. Why Post-Training Does Not Solve It

A natural objection follows: post-training methods such as reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO) exist specifically to correct model errors, so why does hallucination persist — and in some cases worsen — after this stage? The paper's answer is that the technique is not the failure point; the grading is. Prevailing benchmarks score models primarily on accuracy, which means a confident guess is rewarded over an honest expression of uncertainty whenever the guess has any chance of being right. Under that incentive structure, an optimally trained model learns to behave like an overconfident test-taker rather than a calibrated one.

The authors describe this as a socio-technical problem rather than a purely technical one: fixing it requires changing how the field's leaderboards and benchmarks assign credit — specifically, penalizing confident wrong answers more heavily than calibrated abstention, and giving partial credit for an honest “I don't know” — rather than inventing additional hallucination-specific test suites layered on top of an unchanged scoring norm.

4. Corroborating and Complementary Research

This account is consistent with, and extended by, adjacent lines of work. Independent theoretical work has argued that models which are well calibrated on a generative task are mathematically compelled to hallucinate at some rate, since perfect calibration and zero factual error cannot simultaneously hold once rare or singleton facts are present. Detection-oriented work such as SelfCheckGPT approaches the same phenomenon from a black-box angle, showing that sampling a model's own outputs repeatedly and checking for consistency can flag likely-hallucinated spans without needing access to internal model states, which indirectly supports the classification-error framing: inconsistent answers under resampling are a symptom of exactly the kind of uncertainty the theoretical account predicts.

Broader surveys of the field converge on the same conclusion the OpenAI paper reaches from theory: no single mitigation eliminates hallucination, and durable progress requires layered, attribution-aware pipelines rather than any one architectural fix.

5. A Concrete Illustration: Fabricated Citations

The practical stakes of this behavior are visible in academic publishing itself. A 2026 analysis examined 100 fabricated citations that had passed peer review and appeared in accepted NeurIPS 2025 papers, finding they had evaded review by three to five expert researchers per paper and were present in roughly one percent of accepted papers. The same study found more than 50 hallucinated citations in ICLR 2026 submissions under review, some of which had received high reviewer scores, and noted that fabricated citations have also surfaced in government reports and paid consulting deliverables. Citations are a particularly exposed failure surface because they combine precise, low-frequency metadata — author

names, venues, dates — that a model can blend or invent while preserving surface-level plausibility, which lines up directly with the singleton-rate and GIGO mechanisms described in Section 2.

6. Mitigation Strategies

- Retrieval-augmented generation (RAG): grounding generation in retrieved source documents reduces reliance on parametric memory for facts, though it does not eliminate hallucination when retrieval itself fails or returns irrelevant context.
- Calibration and verbalized confidence: training models to report honest confidence levels, so uncertainty is expressed rather than concealed behind fluent prose.
- Evaluation reform: rescore existing benchmarks to reward calibrated abstention and penalize confident errors more than “I don’t know” responses, addressing the incentive structure directly rather than patching outputs after the fact.
- Self-consistency checks: sampling multiple outputs for the same query and flagging low-agreement spans as likely hallucinations.
- Domain-specific fine-tuning and fact-checking pipelines: layering verification stages on top of generation for high-stakes domains such as medicine and law.

7. Takeaways

- Hallucination is a predictable statistical outcome of current training and evaluation norms, not an unexplained defect.
- Some hallucination is mathematically unavoidable for facts seen only once in training, regardless of model scale.
- Post-training reduces certain error classes but cannot fully remove hallucination while benchmarks reward confident guessing over calibrated uncertainty.
- Mitigation is most effective as a layered strategy — retrieval grounding, calibration, and evaluation reform together — rather than any single fix.
- The failure mode is not confined to chatbots: it has already produced fabricated citations in peer-reviewed venues, government reports, and paid professional work.

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